# USB4 2.0 ENGINEERING CHANGE NOTICE FORM

## Title: Remove SBRX_Cin spec limit
## Applied to: USB4 Specification Version 2.0

| **Brief description of the functional changes:** |
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| Remove the SBU Rx input capacitance (SBRX_Cin) limit |

| **Benefits as a result of the changes:** |
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| Remove redundancies in the SBU specifications, while allowing to keep the SBU input capacitance as an internal SBU receiver design parameter without forcing external limit that is not connected to the specific design assumptions that is not impacting the link partner. |

| **An assessment of the impact to the existing revision and systems that currently conform to the USB specification:** |
| --- |
| None |

| **An analysis of the hardware implications:** |
| --- |
| None |

| **An analysis of the software implications:** |
| --- |
| None |

| **An analysis of the compliance testing implications:** |
| --- |
| None |

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## Actual Change

### (a). Table 3-1 in USB4 V1, Table 3-32 in USB4 V2

### From Text:

| Symbol | Description | Min | Max | Units | Conditions |
|---|---|---|---|---|---|
| $SBX_{TRTF}$ | SBTX/SBRX 10-90% Rise/Fall time | 3.5 | 65 | ns | See Note 5. |
| $SBTX_{PULL\_UP\_RES}$ | SBTX pull-up resistor | 7.0K | 10.5K | Ω | See Note 6. |
| $SBRX_{PULL\_DOWN\_RES}$ | SBRX pull-down resistor | 0.70M | 1.05M | Ω | See Note 7. |
| $SBTX_{SOURCE\_IMPEDANCE}$ | SBTX output impedance | 25 | 90 | Ω | |
| SBRX_Cin | SBRX Input Capacitance | -- | 8 | pF | See Note 8. |
| SBX_UI | UI duration | 970 | 1030 | ns | See Note 9. |

Notes:

1. This parameter shall be verified in both transaction and steady state. $SBTX_{VOH}$ and $SBRX_{VIH}$ (min, max1) shall be measured in steady state condition with continuous high level, and (min, max2) shall be measured in transaction mode. Over/undershoot shall be ignored.
2. A buffer may be used between the connector and the SBU transceiver to meet the required voltage levels.
3. Intended to limit the voltage drop on the SBRX input due to current through the SBTX pull-up resistor.
4. Intended to limit the voltage on the SBRX pull-down resistor when the cable is disconnected.
5. Verify this parameter in transaction and not from power down to power up. The minimum is specified to control crosstalk and EMI.
6. A Router shall terminate the SBTX signal to 3.3 V nominal power.
7. A Router shall terminate the SBRX signal to GND.
8. When operating with a Bi-directional Re-timer or linear re-driver in a TBT3 Active Cable, the SBTX will be loaded by 2 * SBRx_Cin. One Cin on the Router SBRX and one in the legacy TBT3 Active Cable or the Linear re-driven Active Cable. See Figure 13-1.
9. SBX_UI corresponds to the instantaneous period of the SBU signal and not to the average period.

### To Text:

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| Symbol | Description | Min | Max | Units | Conditions |
|---|---|---|---|---|---|
| $SBX_{TRTF}$ | SBTX/SBRX 10-90% Rise/Fall time | 3.5 | 65 | ns | See Note 5. |
| $SBTX_{PULL\_UP\_RES}$ | SBTX pull-up resistor | 7.0K | 10.5K | Ω | See Note 6. |
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| $SBTX_{SOURCE\_IMPEDANCE}$ | SBTX output impedance | 25 | 90 | Ω | |
| ~~SBRX_Cin~~ | ~~SBRX Input Capacitance~~ | ~~--~~ | ~~8~~ | ~~pF~~ | ~~See Note 8.~~ |
| SBX_UI | UI duration | 970 | 1030 | ns | See Note ~~9~~8. |

Notes:

1. This parameter shall be verified in both transaction and steady state. $SBTX_{VOH}$ and $SBRX_{VIH}$ (min, max1) shall be measured in steady state condition with continuous high level, and (min, max2) shall be measured in transaction mode. Over/undershoot shall be ignored.
2. A buffer may be used between the connector and the SBU transceiver to meet the required voltage levels.
3. Intended to limit the voltage drop on the SBRX input due to current through the SBTX pull-up resistor.
4. Intended to limit the voltage on the SBRX pull-down resistor when the cable is disconnected.
5. Verify this parameter in transaction and not from power down to power up. The minimum is specified to control crosstalk and EMI.
6. A Router shall terminate the SBTX signal to 3.3 V nominal power.
7. A Router shall terminate the SBRX signal to GND.
8. ~~When operating with a Bi-directional Re-timer or linear re-driver in a TBT3 Active Cable, the SBTX will be loaded by 2 * SBRx_Cin. One Cin on the Router SBRX and one in the legacy TBT3 Active Cable or the Linear re-driven Active Cable. See Figure 13-1.~~

~~9.~~8. SBX_UI corresponds to the instantaneous period of the SBU signal and not to the average period, and shall be measured over the SBTX signal. The SBU receiver design is expected to operate properly with the worst-case incoming SBX_UI and $SBX_{TRTF}$ and while considering the SBU receiver loading.